



COURSE SLIDES

Microsoft 365 Copilot for Excel

Course Code: C34

Conducted by Tertiary Infotech Academy Pte Ltd · UEN 201200696W

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COURSE ADMINISTRATION

Welcome & Housekeeping

Course Trainer Profiles

1

Ken Hiong

Excel practitioner and adult educator with experience in business analysis, planning and reporting.

2

Jim Gan Chiu Liang

Data automation specialist with extensive experience building practical Excel workflows.

3

Liew Sing Loon

Technology and management educator with applied quantitative-analysis experience.

4

Dwight Nuwan Fonseka

Data science leader and adult educator specialising in analytics, machine learning and responsible AI.

Let's Know Each Other

1 Your name, organisation and role.

2 One recurring spreadsheet task that takes too long today.

3 One decision your spreadsheet should help someone make.

4 Whether you can see Copilot in Excel and which app or web version you are using.

Ground Rules

1

Set your mobile phone to silent mode.

2

Participate actively — no question is too small.

3

Mutual respect: agree to disagree.

4

One conversation at a time.

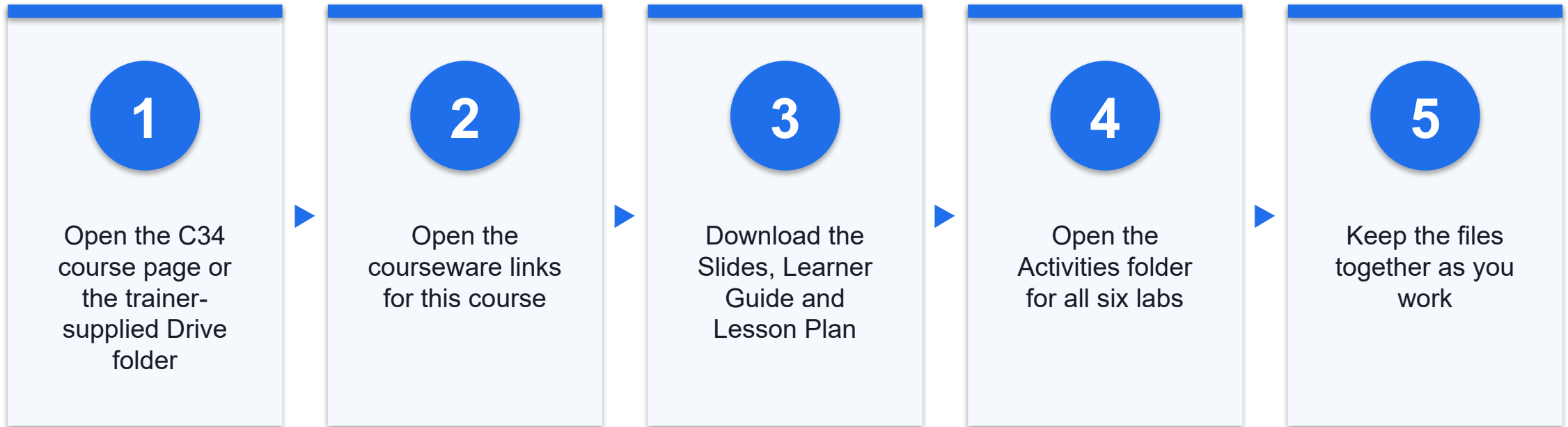
5

Be punctual; return from breaks on time.

6

Learn by doing — try every hands-on lab.

Access Course Materials



Course page: tertiarycourses.com.sg/microsoft-365-copilot-for-excel.html

Lesson Plan — 1 Day, 7.5 instructional hours

Course sequence

- Day 1 — Prepare, transform, analyse and verify one Copilot-assisted Excel workflow
 - Topic 1: Getting Started with Copilot in Excel (Labs 1–3)
 - Topic 2: AI-Powered Data Analysis and Insights (Labs 4–6)

Day format

- Daily timing
 - 9:30 am - 6:30 pm (1-hour lunch; two 15-minute tea breaks; 7.5 instructional hours)
- Instructional time
 - 7.5 hours · 6 connected labs
- Learning arc
 - Concept → demonstrate → build → verify → discuss → recap

Learning Outcomes

1

Prepare & Prompt

Explain how Copilot works in Excel, prepare an analysis-ready Excel table and write bounded prompts for workbook tasks.

2

Transform & Calculate

Use Copilot to highlight, sort and filter table data, then generate and verify auditable formula columns.

3

Summarise & Visualise

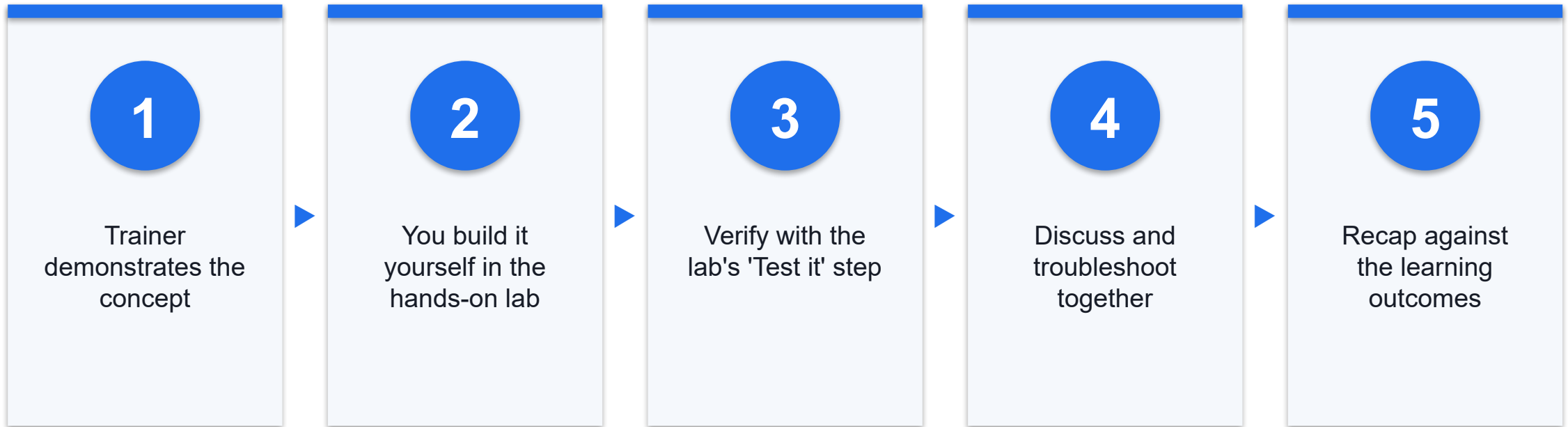
Clean and enrich a business dataset, create PivotTables and charts, and interpret patterns without overstating the evidence.

4

Analyse & Verify

Run advanced analysis with Python-assisted Copilot, reconcile the results and produce a traceable decision summary.

How You'll Learn





CORE CONCEPTS

The Verified Copilot-in-Excel Workflow

Keep Four Layers Visible

1

Source data

Protected raw rows, documented grain, types, identifiers and control totals.

2

Copilot action

The prompt, selected mode, proposed logic and exact workbook change.

3

Verification evidence

Formula checks, reconciliations, Pivot tie-outs and cited source rows.

4

Human decision

Approved wording, limitations, owner and Share, Revise or Hold outcome.

G-C-D-C-O-R Prompt Framework

1

Goal

One task and the decision or workbook change it supports.

2

Context

Audience, business situation, units and time period.

3

Data

Named table, sheets, fields, grain and filters.

4

Constraints

What must remain unchanged, privacy limits and exclusions.

5

Output

Formula, table, chart, explanation or exact schema.

6

Review

Logic, assumptions, edge cases, controls and acceptance criteria.

C34 OPERATING PRINCIPLE

**Copilot accelerates workbook work;
verification determines whether the result is
usable.**

Protect the source, specify the task, inspect every change and reconcile every headline number.

What You Will Build

Prepared workbook

- named Excel table
- readiness log
- prompt-and-change trail

Analysis workbook

- formula columns
- clean table
- PivotTables and charts

Decision pack

- Python-assisted analysis
- reconciliation
- verified executive summary

The Learning Arc in Every Lab

1

Open the supplied checkpoint and state the exact question.

2

Prompt Copilot with a named data boundary and review criteria.

3

Inspect the workbook change, formula, Pivot or generated code.

4

Run the Test It controls and keep the checkpoint for the next lab.

The C34 Verification Gate

1

Structure

Does each row have one meaning and do types and categories match the dictionary?

2

Logic

Do formula, aggregation, denominator, filter and missing-value rules match the question?

3

Reconciliation

Do independent counts and totals tie to the generated output?

4

Visual

Are title, units, scale, categories and source linkage accurate?

5

Claim

Does each sentence distinguish observed evidence from a hypothesis?

6

Decision

Share, Revise or Hold - with an owner, reason and unresolved limitation.

TOPIC 01

Getting Started with Copilot in Excel

Introduction to Microsoft 365 Copilot in Excel · preparing data as Excel tables · effective prompts · highlighting, sorting and filtering · formula columns

Key Concepts — Getting Started with Copilot in Excel

1

Copilot work modes

Choose Edit for workbook changes, Plan for a proposed sequence and Chat for analysis that stays in the pane.

2

Analysis-ready tables

Use one header row, one record per row, one meaning per column and consistent data types.

3

G-C-D-C-O-R prompts

State Goal, Context, Data, Constraints, Output and Review criteria before Copilot acts.

4

Reversible operations

Preview, apply, inspect and undo one bounded change rather than combining many opaque edits.

5

Structured-reference formulas

Prefer formulas that name table columns and update as the table grows.

6

Control totals

Reconcile counts, sums and edge cases independently before using a result in a decision.

Introduction to Microsoft 365 Copilot in Excel

WHAT

- Copilot in Excel is a natural-language interface that can build, edit and analyse workbooks using Excel features such as tables, formulas, PivotTables, charts, formatting and worksheet operations. The current experience offers Edit, Plan and Chat choices. Edit can change the workbook, Plan proposes a sequence before changes and Chat keeps the response in the conversation.

WHY IT MATTERS

- The same prompt has different consequences depending on the mode. A conversational explanation is low-impact; a direct workbook edit changes shared evidence. Choosing the mode deliberately, saving a protected source sheet and keeping version history available makes experimentation reversible and keeps the human owner accountable for the final workbook.

HUMAN CHECK

- Ground the task in approved facts.
- Review evidence, privacy, rights and tone.
- Approve, edit or stop before use.

How It Works — Introduction to Microsoft 365 Copilot in Excel

1

Confirm the eligible Microsoft 365 account, current Excel version and visible Copilot entry point.

2

Choose Chat for questions, Plan when the intended sequence needs review and Edit for an approved change.

3

Inspect the proposed or completed work, save a new version and retain the ability to undo or restore.

In Practice — Introduction to Microsoft 365 Copilot in Excel

WORKED EXAMPLE

- HarbourLight Retail has a raw order export and needs a weekly operations summary.
- The analyst first asks Chat to describe the columns, then asks Plan for a safe preparation sequence.
- Only after reviewing the plan does the analyst use Edit for one bounded change at a time.

DECISION GUIDE

- **Use when**
 - The workbook uses a modern supported format and the required Copilot experience is available.
 - The task has a clear owner who can review changes and restore an earlier version if needed.
- **Avoid when**
 - The prompt would expose secrets, sensitive personal information or data not approved for the service.
 - A high-impact financial, legal or operational decision would rely on an unchecked generated result.

Quality Lens — Introduction to Microsoft 365 Copilot in Excel

Mode

- Match Chat, Plan or Edit to the consequence of the task.

Reversibility

- Protect the source, save versions and make one inspectable change at a time.

Ownership

- A named person reviews and approves the workbook before it is shared or used.

Preparing Data as Excel Tables for Copilot

WHAT

- An analysis-ready table has a single header row, one record per row, one variable per column, stable identifiers and consistent types. Excel tables add names, filters and structured references that expand with new rows. They make the intended data boundary clearer to people, formulas and Copilot.

WHY IT MATTERS

- Merged headings, blank rows, subtotals inside the data region, duplicate identifiers and mixed types create ambiguous grain. If one row sometimes means an order and sometimes a subtotal, no prompt can fully repair the analytical logic. Data structure must be corrected before asking for insight.

HUMAN CHECK

- Ground the task in approved facts.
- Review evidence, privacy, rights and tone.
- Approve, edit or stop before use.

How It Works — Preparing Data as Excel Tables for Copilot

1

Identify the grain and primary identifier, then scan for blank headers, blank rows and duplicated IDs.

2

Convert the bounded range to a table, confirm headers and give the table a descriptive unique name.

3

Check date, text, count, currency and percentage columns; record known defects instead of hiding them.

In Practice — Preparing Data as Excel Tables for Copilot

WORKED EXAMPLE

- Each HarbourLight row represents one order line identified by Order_ID.
- The learner converts the source range into tblOrdersRaw and freezes the header row.
- A data-readiness log records inconsistent Region and Channel text for later controlled cleaning.

DECISION GUIDE

- **Use when**
 - The data can be expressed as a rectangular list with a consistent row meaning.
 - Headers can be made unique and each column can be assigned one stable data type.
- **Avoid when**
 - The range mixes several unrelated tables, presentation titles, subtotals and free-form notes.
 - Duplicate identifiers or missing units make the intended record grain unresolved.

Quality Lens — Preparing Data as Excel Tables for Copilot

Grain

- Write one sentence that says exactly what a row represents.

Schema

- Use unique headers, consistent types and stable categories.

Controls

- Record row count, unique-ID count and source totals before transformation.

Writing Effective Prompts for Data Tasks

WHAT

- A useful prompt is an executable specification. The C34 G-C-D-C-O-R pattern names the Goal, business Context, exact Data boundary, Constraints, required Output and Review checks. It turns words such as 'analyse' or 'clean' into visible acceptance criteria.

WHY IT MATTERS

- Broad prompts force the model to guess which sheet, columns, period, aggregation and format matter. Specific prompts reduce ambiguity and make the result testable. Follow-up prompts should refine one dimension at a time and keep the original analytical question visible.

HUMAN CHECK

- Ground the task in approved facts.
- Review evidence, privacy, rights and tone.
- Approve, edit or stop before use.

How It Works — Writing Effective Prompts for Data Tasks

1

State one goal and the decision or workbook change it should support.

2

Name the table, fields, filters, units and exclusions; request a precise formula, table or chart output.

3

Require Copilot to explain its logic, flag assumptions and show the checks a human should perform.

In Practice — Writing Effective Prompts for Data Tasks

WORKED EXAMPLE

- Goal: identify high-value returned orders that need review.
- Data: tblOrdersRaw; use Returned, Units and Unit_Price; do not infer causes from Customer_Note.
- Output: a filter plus a count; Review: show the exact conditions and preserve the full table.

DECISION GUIDE

- **Use when**
 - The user can name the data boundary and the expected output format.
 - The result has observable acceptance criteria such as a count, formula or explicit condition.
- **Avoid when**
 - The request is 'make this better' with no audience, boundary, rule or decision.
 - The prompt asks Copilot to fill unknown business facts or silently choose a high-impact threshold.

Quality Lens — Writing Effective Prompts for Data Tasks

Specific

- Name the table, fields, period, units and output.

Bounded

- State exclusions, privacy limits and what must remain unchanged.

Reviewable

- Ask for logic, assumptions, control checks and a clear success condition.

Highlighting, Sorting and Filtering Data with Copilot

WHAT

- Highlighting changes visual emphasis, sorting changes visible order and filtering changes which rows are shown. None of them changes the underlying business meaning. A prompt should define the comparison rule, tie-break order, filter conditions and whether hidden rows must remain available.

WHY IT MATTERS

- A visually persuasive sheet can still be analytically wrong. 'Highlight important orders' is undefined; 'highlight Returned=Yes and Gross_Sales at least 1000' is a reproducible rule. Filters can also hide the denominator, so the filtered count and original row count should be recorded together.

HUMAN CHECK

- Ground the task in approved facts.
- Review evidence, privacy, rights and tone.
- Approve, edit or stop before use.

How It Works — Highlighting, Sorting and Filtering Data with Copilot

1

Translate the business question into explicit Boolean conditions and a deterministic sort order.

2

Ask Copilot to describe the intended change, then apply it to the named table.

3

Count visible matches, clear the filter and confirm that the original row count returns.

In Practice — Highlighting, Sorting and Filtering Data with Copilot

WORKED EXAMPLE

- The team reviews orders where Returned is Yes and gross sales are at least S\$1,000.
- Rows are sorted by gross sales descending, then Order_ID ascending to break ties.
- The analyst records both the flagged count and the unchanged total table count.

DECISION GUIDE

- **Use when**
 - The rule can be written using named fields, operators and values.
 - The workbook owner needs a reversible view, not a deletion of non-matching records.
- **Avoid when**
 - Colour alone would carry the only record of a business status.
 - Filtering would hide excluded rows from a report without disclosing the denominator.

Quality Lens — Highlighting, Sorting and Filtering Data with Copilot

Rule

- Write the exact condition before applying any colour or filter.

Denominator

- Record total rows and visible rows together.

Reversible

- Clear the filter and confirm no records were deleted or overwritten.

Generating Formula Columns from Plain-English Requests

WHAT

- Copilot can propose native Excel formulas and add calculated columns. In a table, structured references use names such as [Units] and [Unit_Price], making row logic easier to read and resilient when the table grows. A generated formula remains a hypothesis until it is checked against the business rule.

WHY IT MATTERS

- Formula syntax can be valid while the logic is wrong: discount might be applied twice, margin might use gross rather than net sales, or blanks might become zero without approval. Formula explanation, edge-case tests and independent totals are therefore part of the task, not optional polish.

HUMAN CHECK

- Ground the task in approved facts.
- Review evidence, privacy, rights and tone.
- Approve, edit or stop before use.

How It Works — Generating Formula Columns from Plain-English Requests

1

Write the business rule in words, including units, blank handling, rounding and error behaviour.

2

Ask for one structured-reference formula and an explanation before filling the calculated column.

3

Test representative, boundary and exceptional rows; then reconcile the column to independent totals.

In Practice — Generating Formula Columns from Plain-English Requests

WORKED EXAMPLE

- Gross_Sales equals Units multiplied by Unit_Price.
- Net_Sales equals Gross_Sales less Gross_Sales times Discount_Rate.
- Margin_Rate uses IFERROR(Gross_Profit divided by Net_Sales, zero), formatted as a percentage.

DECISION GUIDE

- **Use when**
 - The rule is deterministic and can be expressed using native Excel functions.
 - A control calculation or small hand-worked sample can verify the result.
- **Avoid when**
 - The requested value is a subjective classification presented as a precise calculation.
 - The formula would embed an unapproved business threshold or conceal missing data.

Quality Lens — Generating Formula Columns from Plain-English Requests

Semantics

- Confirm the numerator, denominator, units, signs and timing.

Edge cases

- Check blanks, zeros, returns, discounts and one high-value row.

Reconciliation

- Tie column totals to an independent control before use.

Hands-On Labs — Getting Started with Copilot in Excel

Lab 1

- Prepare the Workbook and Create the Copilot Control Trail

Lab 2

- Apply Reproducible Highlights, Sorts and Filters

Lab 3

- Generate and Reconcile Formula Columns

Prepare the Workbook and Create the Copilot Control Trail

LAB 1

You will import the synthetic HarbourLight order file, save it as a modern Excel workbook, convert the source rows into a named table and create Control and Prompt_Log sheets. You will use Copilot in Chat and Plan only to describe the data and propose a safe sequence. The raw source remains unchanged.

You'll build

work-c34/C34-HarbourLight-Copilot-Analysis.xlsx with Raw_Orders, Control and Prompt_Log sheets, a tblOrdersRaw table, baseline counts and a recorded Copilot access note

1 Save the source

2 Define the grain

3 Create the table

4 Record controls

Tools: Excel for Microsoft 365 or Excel for the web · Copilot · harbourlight-orders-raw.csv

Prepare the Workbook and Create the Copilot Control Trail

Test it

Raw_Orders must contain table tblOrdersRaw with exactly 36 data rows and 36 populated Order_ID values. Control must show earliest date 2026-01-05, latest date 2026-04-28, 36 rows, 36 populated IDs and the same Units and gross-sales controls as labs/assets/expected-controls.md. Prompt_Log must contain one Chat and one Plan record, both with 'No change' in Workbook_Change. Raw_Orders must still match the CSV.

Apply Reproducible Highlights, Sorts and Filters

LAB 2

You will copy the source into a working sheet, state deterministic review rules and ask Copilot to apply highlighting, sorting and filtering one operation at a time. You will record total and visible row counts, clear all filters and confirm that the source and working tables still contain the same orders.

You'll build

An Orders_Working sheet with tblOrdersWorking, reversible highlighting, an explicit multi-level sort, a returned-order filter and Prompt_Log evidence for every applied change

1 Copy the source

2 Write the rules

3 Apply one change

4 Count matches

Tools: Excel · Copilot Edit or Plan · Control sheet · Prompt_Log

Apply Reproducible Highlights, Sorts and Filters

Test it

The working table must contain 36 rows before and after the filter. The returned-order filter must show the exact count in labs/assets/expected-controls.md; the high-discount rule must match the listed count; and the maximum Units value plus first tie-break order must match the control. After Data > Clear, all 36 orders must be visible. Raw_Orders values and order must remain unchanged.

Generate and Reconcile Formula Columns

LAB 3

You will write the calculation rules in words, ask Copilot for one structured-reference formula at a time and inspect every formula before filling it through the table. You will verify a no-discount row, a discounted row, a returned order, the zero-error path and independent totals on the Control sheet.

You'll build

Six calculated columns in tblOrdersWorking - Gross_Sales, Discount_Amount, Net_Sales, Cost_Amount, Gross_Profit and Margin_Rate - plus formula checks and reconciled totals on Control

1 Define the rules

2 Generate one formula

3 Inspect the fill

4 Test edge cases

Tools: Excel formulas · Copilot · structured references · expected-controls.md

Generate and Reconcile Formula Columns

Test it

All six calculated columns must contain formulas for all 36 rows with no formula-error values. The formula text must use structured references and the overall margin must use total Gross_Profit divided by total Net_Sales. The six totals and the named representative Order_ID checks must exactly match labs/assets/expected-controls.md. Copilot's explanation alone does not satisfy the check.

Recap - Getting Started with Copilot in Excel

1

LO1 - Prepare and Prompt: Confirm the available Copilot experience, keep a protected source sheet, convert one-row-per-order data into a named Excel table and use Goal-Context-Data-Constraints-Output-Review prompts.

2

LO2 - Transform and Calculate: Apply explicit highlight, sort and filter rules; generate structured-reference formulas; and reconcile row-level calculations to independent control totals before relying on them.

TOPIC 02

AI-Powered Data Analysis and Insights

Cleaning and enriching data · PivotTables and charts from a prompt · trends, outliers and insights · advanced analysis with Python in Excel · verification and prompt best practices

Key Concepts — AI-Powered Data Analysis and Insights

1

Defined cleaning rules

Correct only documented text, spacing, format and missing-value issues; preserve the raw source.

2

Question-to-visual design

Choose grain, dimensions, measures, aggregation and chart form from the decision question.

3

Descriptive insight

Describe comparison, trend, distribution and outlier evidence before considering explanations.

4

Python-assisted analysis

Review the referenced data, generated code, statistical method and returned output together.

5

Static versus refreshable output

Label inserted images or tables that do not update when source data changes.

6

Evidence chain

Trace every headline claim to source rows, a calculation and an independent cross-check.

Cleaning and Enriching Data with Copilot

WHAT

- Cleaning makes existing values conform to a defined schema; enrichment adds a new field from a rule, lookup or reviewed classification. The source sheet should remain unchanged. The cleaning log records the issue, rule, affected count and verification so corrections are auditable.

WHY IT MATTERS

- Inconsistent text, number formats and extra spaces can split categories or produce wrong summaries. Microsoft's Clean Data experience focuses on these common patterns, but availability can vary. A manual or Copilot-assisted correction still needs a canonical value list and before/after counts.

HUMAN CHECK

- Ground the task in approved facts.
- Review evidence, privacy, rights and tone.
- Approve, edit or stop before use.

How It Works — Cleaning and Enriching Data with Copilot

1

Profile the source and compare observed values with the supplied data dictionary.

2

Copy the table to a working sheet; correct one issue type at a time and log the affected rows.

3

Add deterministic enrichment first, then review any text classification row by row against a taxonomy.

In Practice — Cleaning and Enriching Data with Copilot

WORKED EXAMPLE

- Region values such as 'north', 'North ' and 'NORTH' become the canonical value 'North'.
- Channel values map only to Store, Online or Partner according to the dictionary.
- Customer_Note receives a reviewed Note_Theme; blanks remain 'No note' rather than invented feedback.

DECISION GUIDE

- **Use when**
 - The canonical schema and acceptable value set are documented.
 - The original data and a change log are retained for comparison.
- **Avoid when**
 - A correction would guess a missing business fact or overwrite an identifier.
 - A free-text classification will be treated as objective truth without human review.

Quality Lens — Cleaning and Enriching Data with Copilot

Preserve

- Keep the raw sheet unchanged and make corrections in a working table.

Count

- Record affected rows before and after each rule.

Taxonomy

- Use a small defined category list and retain Unknown when evidence is insufficient.

Building PivotTables and Charts from a Prompt

WHAT

- A PivotTable aggregates measures across dimensions at a chosen grain. A chart encodes that summary visually. A complete prompt names the source table, row and column fields, aggregation, filters, order, chart type, title and the decision the view should support.

WHY IT MATTERS

- Copilot can build a polished result quickly, but speed does not resolve analytical design. Sum of sales, average margin and count of orders answer different questions. A chart can also exaggerate differences through an unsuitable scale, missing units or too many categories.

HUMAN CHECK

- Ground the task in approved facts.
- Review evidence, privacy, rights and tone.
- Approve, edit or stop before use.

How It Works — Building PivotTables and Charts from a Prompt

1

State the decision question, grain, dimensions, measures and required filters.

2

Generate the PivotTable first and reconcile its grand total to the cleaned table.

3

Choose a chart that fits comparison or time trend, then verify title, units, labels and source link.

In Practice — Building PivotTables and Charts from a Prompt

WORKED EXAMPLE

- Question: how did monthly net sales change by canonical region?
- Rows use Month, columns use Region and values use Sum of Net_Sales.
- A line chart shows month on the x-axis and S\$ net sales on the y-axis; the Pivot grand total must tie out.

DECISION GUIDE

- **Use when**
 - The table has a stable grain and measures have defined aggregations.
 - The intended comparison or trend can be shown without hiding material categories.
- **Avoid when**
 - The measure is a percentage that should be recomputed from totals rather than averaged row by row.
 - A chart is requested before the question, denominator or aggregation is agreed.

Quality Lens — Building PivotTables and Charts from a Prompt

Grain

- Name what one Pivot cell represents.

Tie-out

- Reconcile the Pivot grand total to the source-table control.

Chart truth

- Show units, full categories and an honest scale appropriate to the question.

Surfacing Trends, Outliers and Insights

WHAT

- A trend describes change across an ordered period; an outlier is an observation unusually distant under a stated rule; an insight connects a verified pattern to a relevant decision. None of these, by itself, proves why the pattern occurred.

WHY IT MATTERS

- Prompts such as 'what is interesting?' can surface useful candidates, but they can also produce selective or causal-sounding narratives. Start with a baseline, comparison and explicit outlier rule. Label observed values separately from hypotheses that require more evidence.

HUMAN CHECK

- Ground the task in approved facts.
- Review evidence, privacy, rights and tone.
- Approve, edit or stop before use.

How It Works — Surfacing Trends, Outliers and Insights

1

Define the metric, period, comparison and minimum volume needed for interpretation.

2

Ask for descriptive results and an explicit outlier rule before requesting explanations.

3

Cross-check the cited rows, record limitations and convert only relevant verified patterns into actions.

In Practice — Surfacing Trends, Outliers and Insights

WORKED EXAMPLE

- Monthly net sales are compared by region, while returned-order rate uses returned orders divided by orders.
- High-value orders are flagged by a stated IQR rule and then inspected for data-quality issues.
- A spike is described as observed; promotion, channel mix or data error remain hypotheses.

DECISION GUIDE

- **Use when**
 - The time order, denominator and comparison group are stable.
 - The output cites underlying values and states the method used to identify an outlier.
- **Avoid when**
 - A short or incomplete period is used to claim a persistent trend.
 - Correlation or a coincident change is presented as a proven cause.

Quality Lens — Surfacing Trends, Outliers and Insights

Baseline

- Show the comparison period, denominator and sample size.

Method

- State the trend or outlier rule in reproducible terms.

Language

- Use observed for evidence and hypothesis for a possible explanation.

Advanced Analysis with Python in Excel

WHAT

- Copilot can use Python-based analysis to interpret a question, compute statistics and return explanations, tables or visuals. Direct-answer output may be static. Advanced analysis can open a new sheet and expose Python code or a refreshable Python cell, depending on the available experience.

WHY IT MATTERS

- Python expands the range of analysis but does not remove review obligations. The analyst must verify the referenced data, cleaning choices, grouping logic, missing-value treatment, statistical method and returned values. Python in Excel runs in a Microsoft cloud container and has defined access and network boundaries.

HUMAN CHECK

- Ground the task in approved facts.
- Review evidence, privacy, rights and tone.
- Approve, edit or stop before use.

How It Works — Advanced Analysis with Python in Excel

1

Ask a precise analytical question and state the table, fields, period, grouping and expected output.

2

Open the generated logic or code; inspect row filters, aggregations, missing values and method assumptions.

3

Recompute headline results with Excel or a PivotTable, label static outputs and save the analysis notes.

In Practice — Advanced Analysis with Python in Excel

WORKED EXAMPLE

- The team asks for monthly net-sales trends, return rates and an IQR review of high-value orders.
- Generated Python groups only canonical rows and reports the count excluded for unresolved categories.
- Excel control cells and the Pivot grand total independently confirm the headline values.

DECISION GUIDE

- **Use when**
 - The question needs deeper exploratory statistics or visuals and the available license supports the feature.
 - A reviewer can inspect the generated method and reproduce headline results independently.
- **Avoid when**
 - The analysis depends on unavailable external network calls or hidden local files.
 - A static inserted output will be mistaken for a refreshable result after the source changes.

Quality Lens — Advanced Analysis with Python in Excel

Inputs

- Confirm the exact table, rows, columns, filters and missing-value treatment.

Method

- Review code and assumptions; do not accept the narrative alone.

Reproducibility

- Tie headline values to Excel controls and label static outputs visibly.

Verifying Results and Prompt Best Practices

WHAT

- Verification is a chain of checks from source structure to formula, aggregation, visual and written claim. The C34 quality gate uses protected source data, a prompt log, edge-case tests, independent control totals, traceable claims and a named human decision.

WHY IT MATTERS

- Copilot can produce fluent explanations and valid-looking formulas that are incomplete or wrong. Microsoft advises users to review, edit and verify generated content. A compact repeatable gate makes that advice operational and prevents an attractive workbook from bypassing basic evidence checks.

HUMAN CHECK

- Ground the task in approved facts.
- Review evidence, privacy, rights and tone.
- Approve, edit or stop before use.

How It Works — Verifying Results and Prompt Best Practices

1

Validate row grain, types, category values, duplicates and source control totals.

2

Inspect formulas, representative rows, Pivot totals, chart encodings and Python logic.

3

Record each claim, supporting value, limitation, owner and Share, Revise or Hold decision.

In Practice — Verifying Results and Prompt Best Practices

WORKED EXAMPLE

- The executive summary states a regional sales movement and cites the Pivot values behind it.
- A return-rate statement shows numerator and denominator, not only the percentage.
- An outlier explanation stays a hypothesis until operational evidence supports it.

DECISION GUIDE

- **Use when**
 - Any generated output will be shared, used in a decision or reused as a template.
 - Another reviewer needs to reproduce the conclusion without relying on the chat history.
- **Avoid when**
 - A fluent answer is treated as proof without checking the workbook.
 - A control total is copied from the same generated output it is supposed to verify.

Quality Lens — Verifying Results and Prompt Best Practices

Independent

- Use a different method or source for the control check.

Traceable

- Link every headline statement to cells, rows or a documented calculation.

Decisive

- Choose Share, Revise or Hold and state the owner and unresolved limitation.

Hands-On Labs — AI-Powered Data Analysis and Insights

Lab 4

- Clean and Enrich the Analysis Table

Lab 5

- Build Reconciled PivotTables and Charts

Lab 6

- Run Advanced Analysis and Write the Verified Decision Summary

Clean and Enrich the Analysis Table

LAB 4

You will copy the formula-complete working table to Orders_Clean, profile observed category values and apply the supplied canonical rules one issue at a time. You will add Month, Return_Flag and Note_Theme fields, retain Unknown when a note is ambiguous and prove that cleaning did not change row-level monetary results.

You'll build

An Orders_Clean sheet with tblOrdersClean, canonical categories, Month, Return_Flag and Note_Theme fields, plus a cleaning log and before/after control checks

1 Copy the working table

2 Profile categories

3 Clean one rule

4 Add analysis fields

Tools: Excel · Copilot · Clean Data when available · harbourlight-data-dictionary.md

Clean and Enrich the Analysis Table

Test it

tblOrdersClean must contain 36 rows and 36 populated Order_ID values. Region must have exactly three canonical values and Channel exactly three. Month must contain only 2026-01 through 2026-04. Sum of Return_Flag, the five monetary totals and all Note_Theme counts must match labs/assets/expected-controls.md. Raw_Orders and Orders_Working must remain unchanged.

Build Reconciled PivotTables and Charts

LAB 5

You will define two decision questions before asking Copilot to build anything. The first PivotTable and line chart show monthly net sales by canonical region. The second summary shows returned orders, order count and return rate by channel. You will verify grand totals, denominators, titles, units and source linkage.

You'll build

A Pivot_Analysis sheet with two PivotTables, a monthly regional net-sales line chart, a channel return-rate chart, tie-out cells and a Prompt_Log record for each object

1 State the questions

2 Build the Pivot

3 Tie the total

4 Create the chart

Tools: Excel PivotTables · Excel charts · Copilot · tblOrdersClean · expected-controls.md

Build Reconciled PivotTables and Charts

Test it

Pivot A must contain four months, three regions and a grand total equal to tblOrdersClean Net_Sales with a 0.00 tie-out. Pivot B must show three channels, returned-order count, order count and rates that match labs/assets/expected-controls.md; the overall rate must use grand-total numerator and denominator. Both charts must be editable, linked, fully titled and use the correct units.

Run Advanced Analysis and Write the Verified Decision Summary

LAB 6

You will ask Copilot for direct answers and advanced analysis using only `tblOrdersClean`. You will review the referenced fields, filters, groupings, IQR logic and generated code or explanation. You will reconcile the headline values to the Control and Pivot_Analysis sheets, then create a concise executive summary that keeps observations, hypotheses and limitations separate.

You'll build

An Advanced_Analysis sheet with prompt, method, generated output and reconciliation; plus an Executive_Summary sheet with three verified findings, limitations, actions and a final decision

1 Write the brief

2 Run advanced analysis

3 Inspect the logic

4 Reconcile headlines

Tools: Copilot direct answers · advanced analysis with Python in Excel · Excel controls · Pivot_Analysis · advanced-analysis-fallback.md

Run Advanced Analysis and Write the Verified Decision Summary

Test it

Advanced_Analysis must contain the completed brief, direct answer, reviewed code or method, output-type label and reconciliation table. Every required metric and outlier must match labs/assets/expected-controls.md within the stated tolerance. Executive_Summary must contain exactly three findings, evidence references, types, next checks, owners, four required limitations and a final SHARE, REVISE or HOLD decision. SHARE is allowed only when every reconciliation status is OK.

Recap - AI-Powered Data Analysis and Insights

1

LO3 - Summarise and Visualise: Correct defined quality issues, build PivotTables at the intended grain, choose charts that match the question and separate observed patterns from possible explanations.

2

LO4 - Analyse and Verify: State the analytical question, review Python-based logic and outputs, cross-check headline values, record limitations and communicate only decision-relevant, evidence-backed findings.



WRAP-UP

Course Summary & Next Steps

What You Achieved

1

Prepare & Prompt

Explain how Copilot works in Excel, prepare an analysis-ready Excel table and write bounded prompts for workbook tasks.

2

Transform & Calculate

Use Copilot to highlight, sort and filter table data, then generate and verify auditable formula columns.

3

Summarise & Visualise

Clean and enrich a business dataset, create PivotTables and charts, and interpret patterns without overstating the evidence.

4

Analyse & Verify

Run advanced analysis with Python-assisted Copilot, reconcile the results and produce a traceable decision summary.

Continue the Verified Workflow

1

Reuse the prepared prompt and verification templates on an authorised workbook.

2

Keep a protected source sheet and visible controls in every Copilot-assisted model.

3

Refresh PivotTables and rerun checks whenever the source data changes.

4

Review generated code, formulas and claims before sharing a decision summary.

Keep Practising After the Course

1

Re-run the labs on your own machine — repetition is what makes the skills stick.

2

Apply the techniques to a real process or project in your own organisation.

3

Keep your downloaded C34 course files and use the lab checkpoints to restart safely.

4

Explore the follow-on courses at www.tertiarycourses.com.sg.

Thank You!

You can now prepare, transform, analyse and verify Excel work with Copilot while keeping source evidence, formula logic and human judgment in control.